

PERIODIC SHADOWING FOR BILATERAL WEIGHTED SHIFTS IS THE SAME ON $\ell_p(\mathbb{Z})$ AND $c_0(\mathbb{Z})$

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ABSTRACT. We answer Question 4.13 of Hevessy–Raunig’s paper *Dynamical properties of weighted shifts on sequence spaces*. Let $w = (w_n)_{n \in \mathbb{Z}}$ be bounded and bounded away from zero, and let B_w be the bilateral weighted backward shift. Then, for every $1 \leq p < \infty$, B_w has the periodic shadowing property on $\ell_p(\mathbb{Z})$ if and only if it has the periodic shadowing property on $c_0(\mathbb{Z})$. The same answer holds for bilateral weighted forward shifts.

1. SOURCE QUESTION

The source paper studies the bilateral weighted backward shift

$$B_w(x)_j = w_{j+1}x_{j+1}$$

on $\ell_p(\mathbb{Z})$, $1 \leq p < \infty$, and on $c_0(\mathbb{Z})$, where w is bounded and $\inf_j |w_j| > 0$. After proving a complete c_0 -characterization of the periodic shadowing property, the authors ask:

Suppose this is not the case. Then we have $|y_r(l+r)| = \|y_r\| \geq \delta > |x_0(l+r)|$. Note that we have $|y_0(l)| \leq |x_r(l)|$ by the definition of y_0 . Let $u < r$ be maximal such that

$$(2) \quad |y_u(l+u)| \leq |x_{r-u}(l+u)|$$

(mind you, we have just shown that this holds for $u = 0$, but not $u = r$).

By the assumption, we have that $\delta \leq |y_u(l+u)|$. This and the definition of y_i implies that

$$(3) \quad |B_w(y_{u+1})(l+u)| + \delta = |y_u(l+u)|.$$

Since $(x_i)_{i=0}^n$ is a δ -pseudotrajectory we have that

$$(4) \quad |(B_w(x_{r-u-1}) - x_{r-u})(l+u)| < \delta.$$

All together we get

$$|B_w(y_{u+1})(l+u)| \stackrel{(3)}{=} |y_u(l+u)| - \delta \stackrel{(2)}{\leq} |x_{r-u}(l+u)| - \delta \stackrel{(4)}{\leq} |B_w(x_{r-u-1})(l+u)|.$$

But since B_w^{-1} preserves the pointwise order we get that $|y_{u+1}(l+u+1)| \leq |x_{r-u-1}(l+u+1)|$, which contradicts the maximality of u . Thus the claim is proved.

Let p be such that $\|y_p\| < \delta$. Using simple induction and (1), one can show that

$$\delta > \|y_p\| = |y_p(l+p)| = \left| \frac{\varepsilon}{w(l+1) \dots w(l+p)} - \delta \sum_{i=0}^{1-p} \frac{1}{w(l+p) \dots w(l+p+j)} \right|.$$

By taking $l+p = m$ we get the first part of the claim. The second part can be shown similarly by starting with $z_0 = B_w(y_0)$ and doing similar construction using B_w instead of B_w^{-1} . $\square$

The curious mind may wonder if one can obtain the same equivalence for ℓ_p . Instead we pose the following question which is easily seen to be equivalent to asking if our characterisation holds for ℓ_p .

Question 4.13. Does the periodic shadowing property for weighted backward (equiv. forward) shifts on $\ell_p(\mathbb{Z})$ ($1 \leq p < \infty$) coincide with the periodic shadowing property on $c_0(\mathbb{Z})$?

Note that the shadowing property for weighted backward shifts on $\ell_p(\mathbb{Z})$ ($1 \leq p < \infty$) coincides with the shadowing property on $c_0(\mathbb{Z})$ [5, Theorem 18] and also the chain recurrence for weighted backward shifts for $\ell_p(\mathbb{Z})$ ($1 \leq p < \infty$) coincides with chain recurrence for $c_0(\mathbb{Z})$ [6, Theorem 14]. Since, by Theorem 4.7 the periodic shadowing property is in between those two properties, a negative answer to the question would be surprising. However, there are some properties of weighted shifts that differ on those spaces, see e.g. [9].

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In the numbering of the rendered arXiv v2 PDF this is Question 4.13: does periodic shadowing for weighted backward, equivalently forward, shifts on $\ell_p(\mathbb{Z})$, $1 \leq p < \infty$, coincide with periodic shadowing on $c_0(\mathbb{Z})$?

2. PROOF INTUITION

The only obstruction in the source paper is the case in which the shift has no nonzero periodic points. In that case, periodic shadowing means that every sufficiently accurate periodic pseudo-orbit must be small, since the only periodic point available to shadow it is 0.

For a period N pseudo-orbit $x = (x_0, \dots, x_{N-1})$, write its cyclic defect as

$$(D_N x)_i = x_{i+1} - B_w x_i, \quad i \in \mathbb{Z}/N\mathbb{Z}.$$

Thus periodic shadowing with no nonzero periodic points is the uniform estimate

$$\max_i \|x_i\| \leq C \max_i \|x_{i+1} - B_w x_i\|$$

for all periods N .

For c_0 , this estimate is scalar: the operator D_N decomposes along the diagonal orbits of the space-time lattice $(i, j) \mapsto (i+1, j-1)$. The c_0 estimate is exactly a uniform bound on the scalar Green matrices for these first-order diagonal recurrences. Those matrices have bounded row sums; applying the same argument to the inverse shift gives the corresponding column control. The Schur test then lifts the scalar estimate to the mixed norm $\max_i \|x_i\|_p$. This removes the only missing implication in the source paper.

3. PRELIMINARIES

We use three facts from the source paper [1].

- (1) The periodic points of B_w are either trivial or dense on each of the spaces under consideration; if they are dense, then periodic shadowing is equivalent to the ordinary shadowing property [1, Proposition 4.5].
- (2) The ordinary shadowing property for these weighted shifts is the same on $\ell_p(\mathbb{Z})$, $1 \leq p < \infty$, and on $c_0(\mathbb{Z})$; the source cites this as Bernardes–Messaoudi’s Theorem 18 [2].
- (3) If $\text{Per}(B_w) = \{0\}$, periodic shadowing is equivalent to the following homogeneous cyclic estimate: for every period N there is one constant C , independent of N , such that every N -periodic pseudo-orbit satisfies

$$\max_{0 \leq i < N} \|x_i\| \leq C \max_{0 \leq i < N} \|x_{i+1} - B_w x_i\|.$$

This is Proposition 4.8 of the source paper, with the usual scaling argument.

We record the only new ingredient separately.

Lemma 1 (Regular Green-matrix transference). *Assume that B_w has no nonzero periodic points on $c_0(\mathbb{Z})$. If the cyclic defect estimate above holds on $c_0(\mathbb{Z})$, then it holds on $\ell_p(\mathbb{Z})$ for every $1 \leq p < \infty$.*

Proof. Fix a period N . For $x = (x_i)_{i \in \mathbb{Z}/N\mathbb{Z}}$, put $e = D_N x$, so

$$e_i(j) = x_{i+1}(j) - w_{j+1} x_i(j+1).$$

For each residue $r \in \mathbb{Z}/N\mathbb{Z}$, follow the diagonal $i + j = r$ and define

$$a_j^{(r)} = x_{r-j}(j), \quad b_j^{(r)} = e_{r-j-1}(j),$$

with time indices understood modulo N . Then

$$(1) \quad a_j^{(r)} - w_{j+1} a_{j+1}^{(r)} = b_j^{(r)}.$$

Thus D_N is a direct sum, after this diagonal reindexing, of scalar first-order difference equations. The c_0 cyclic estimate says that the solution operator for (1), restricted to c_0 -solutions, is bounded in sup norm by a constant C independent of N and r .

Apply this estimate first to B_w , and then to B_w^{-1} . Periodic shadowing is invariant under inversion: reversing a periodic pseudo-orbit of B_w gives a periodic pseudo-orbit of B_w^{-1} , with constants changed only by the fixed bounds on B_w and B_w^{-1} . Consequently the finite Green matrices G_N associated with (1) have uniformly bounded regular Schur norm: there is M , independent of N , such that for every finite coordinate truncation of the diagonal equations,

$$\sup_{\alpha} \sum_{\beta} |(G_N)_{\alpha\beta}| \leq M, \quad \sup_{\beta} \sum_{\alpha: t(\alpha)=t_0} |(G_N)_{\alpha\beta}| \leq M$$

for every output time t_0 . The first bound is the sup-norm estimate for arbitrary choices of defect phases; the second is the same row-sum estimate for the reversed cyclic equation, rewritten as a column estimate for a fixed output time.

Now let $x_i \in \ell_p(\mathbb{Z})$ and $e = D_N x$. By density it is enough to prove the estimate for finitely supported arrays and then pass to the limit. Write the Green representation with nonnegative coefficients $g_{\alpha\beta} = |(G_N)_{\alpha\beta}|$:

$$|x_{t_0}(j)| \leq \sum_{\beta} g_{(t_0,j),\beta} |e_{\beta}|.$$

Using Jensen's inequality with the row-sum bound and then the time-resolved column bound,

$$\begin{aligned} \|x_{t_0}\|_p^p &\leq M^{p-1} \sum_j \sum_{\beta} g_{(t_0,j),\beta} |e_{\beta}|^p \\ &\leq M^p \max_{0 \leq i < N} \|e_i\|_p^p. \end{aligned}$$

Taking p -th roots and then the supremum over t_0 gives

$$\max_i \|x_i\|_p \leq M \max_i \|(D_N x)_i\|_p,$$

with M independent of N . This is exactly the cyclic defect estimate on $\ell_p(\mathbb{Z})$. $\square$

4. MAIN RESULT

Theorem 1. *Let $w = (w_n)_{n \in \mathbb{Z}}$ be bounded and satisfy $\inf_n |w_n| > 0$. Let B_w be the bilateral weighted backward shift on $c_0(\mathbb{Z})$ and on $\ell_p(\mathbb{Z})$, $1 \leq p < \infty$. Then*

$$\begin{aligned} &B_w \text{ has the periodic shadowing property on } c_0(\mathbb{Z}) \\ &\iff B_w \text{ has the periodic shadowing property on } \ell_p(\mathbb{Z}). \end{aligned}$$

Proof. First suppose that B_w has the periodic shadowing property on $c_0(\mathbb{Z})$.

If B_w has a nonzero periodic point on $c_0(\mathbb{Z})$, then the periodic points are dense and the source paper gives periodic shadowing $\iff$ ordinary shadowing. Since ordinary shadowing is the same on $c_0(\mathbb{Z})$ and $\ell_p(\mathbb{Z})$, B_w has shadowing, hence periodic shadowing, on $\ell_p(\mathbb{Z})$.

If B_w has no nonzero periodic point on $c_0(\mathbb{Z})$, then it also has no nonzero periodic point on $\ell_p(\mathbb{Z})$, because $\ell_p(\mathbb{Z}) \subset c_0(\mathbb{Z})$. Proposition 4.8 of the source paper converts c_0 -periodic shadowing into the uniform cyclic defect estimate on c_0 . Lemma 1 transfers that estimate to ℓ_p . Applying Proposition 4.8 again gives periodic shadowing on $\ell_p(\mathbb{Z})$.

Conversely suppose that B_w has periodic shadowing on $\ell_p(\mathbb{Z})$.

If $\text{Per}(B_w)$ is nontrivial on $\ell_p(\mathbb{Z})$, then it is dense, and the same argument as above gives ordinary shadowing on $\ell_p(\mathbb{Z})$, hence ordinary shadowing and periodic shadowing on $c_0(\mathbb{Z})$.

It remains to consider the case $\text{Per}(B_w) = \{0\}$ on $\ell_p(\mathbb{Z})$. The source paper proves that, in this case, periodic shadowing on $\ell_p(\mathbb{Z})$ implies the scalar weight condition appearing in its Theorem 4.12. If $c_0(\mathbb{Z})$ had a nonzero periodic point, then c_0 -chain recurrence would be nontrivial; but chain

recurrence for these shifts is the same on $\ell_p(\mathbb{Z})$ and $c_0(\mathbb{Z})$ by Bernardes–Peris [3, Theorem 14], as recalled in the source paper after Question 4.13. This would contradict the source observation that periodic shadowing with no nonzero periodic points forces the chain recurrent set to be $\{0\}$. Thus $\text{Per}(B_w) = \{0\}$ on $c_0(\mathbb{Z})$. The complete c_0 -characterization in Theorem 4.12 of the source paper now converts the same scalar weight condition into periodic shadowing on $c_0(\mathbb{Z})$. $\square$

Corollary 1. *The same equivalence holds for bilateral weighted forward shifts.*

Proof. If F_w is invertible, then F_w^{-1} is a weighted backward shift with reciprocal shifted weights. Periodic shadowing is invariant under inversion, so Theorem 1 applies to F_w^{-1} and hence to F_w . $\square$

5. VERIFICATION AND NOVELTY NOTES

The proof uses the source paper’s results for the dense-periodic-points case, the source paper’s reduction of the zero-periodic-points case to cyclic pseudo-orbit estimates, and the new regular Green-matrix transference lemma above.

The verifier script in the packet’s `code/` folder checks the finite cyclic model behind Lemma 1: after diagonal reindexing, the cyclic defect operator splits into first-order scalar blocks, and the sampled mixed ℓ_p amplification is bounded by the Schur constant computed from the finite Green matrix. This is only a sanity check of the finite algebra; the proof of Lemma 1 is the mathematical argument.

Local run indexes were searched for arXiv:2509.11852, the paper title, “periodic shadowing”, “weighted backward shifts”, “ell_p”, and “c_0”. No duplicate settlement was found. Web searches for the exact question and close variants found the source arXiv page, a seminar announcement, a ResearchGate mirror, and summary/review pages, but no later paper answering Question 4.13.

Human review should focus on Lemma 1, specifically the derivation of the time-resolved column bound from the reversed cyclic equation.

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